

General, Course structure & Theme

Course code and definition:

Course code	Definitions
L	Lecture
T	Tutorial
P	Practical
BSC	Basic Science Courses
ESC	Engineering Science Courses
HSMC	Humanities and Social Sciences including Management courses
PCC	Professional core courses
PEC	Professional Elective courses
OEC	Open Elective courses

Definition of Credit:

1 Hr. Lecture (L) per week	1 credit
1 Hr. Tutorial (T) per week	1 credit
1 Hr. Practical (P) per week	0.5 credits
2 Hours Practical (Lab)/week	1 credit

Range of credits:

A range of credits from 150 to 160 is required for a student to be eligible to get Under Graduate degree in Engineering. A student will be eligible to get Under Graduate degree with Honours or additional Minor Engineering, if he/she completes an additional 20 credits. These could be acquired through MOOCs.

Structure of Undergraduate Engineering program:

Category Index	Category name	Ideal credit set by AICTE	Actual credit proposed
HSMC	Humanities and social science including management	12	9
BSC	Basic Science course	25	23
ESC	Engineering Science course	24	25.5
PCC	Professional core courses	48	57
PEC	Professional elective courses	18	15
OEC	Open elective course	18	18
PROJ	Project work, seminar and internship in industry or elsewhere	15	13

MC	Mandatory courses	Non credit	No credit
Total		160	160.5

I. Semester-wise structure of curriculum
[L= Lecture, T = Tutorials, P = Practical & C = Credits]

Semester I (First year]
Branch/Course Mechanical Engineering

Sl no	Category Index	Paper Code	Name of the Paper	Periods per week				Credit
				L	T	P	Total	
1	ESC	MENUGES01	Engineering Mechanics	3	1	0	4	4
2	ESC	ECEUGES01	Basic Electronics Engineering	3	0	0	3	3
3	BSC	MATUGBS01	Engineering Mathematics I	3	1	0	4	4
4	BSC	PHYUGBS01	Engineering Physics	3	0	0	3	3
5	ESC	CENUGES01	Engineering Drawing	1	0	3	4	2.5
6	ESC	ECEUGES02	Basic Electronics Engineering Lab	0	0	3	3	1.5
7	BSC	PHYUGBS02	Engineering Physics lab	0	0	3	3	1.5
8	MC	UCCUGAU01	Arabic language & Islamic Studies	4	0	0	4	0
Total credits							19.5	
Total Period per week							28	

** Students will undergo an induction program of 3 weeks duration during the First Semester

Detailed syllabus

Semester I (First year)

ESC	MENUGES 01	Engineering Mechanics	3L-1T-0P	4 Credits
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Course Outcomes: Upon completion of this course, students will be able to grasp the following concepts –

- Use scalar and vector analytical techniques for analyzing forces in statically determinate structures.
- Apply fundamental concepts of kinematics and kinetics of particles to the analysis of simple, practical problems.
- Understand basic kinematics concepts – displacement, velocity and acceleration (and their angular counterparts);
- Understand basic dynamics concepts – force, momentum, work and energy;
- Understand and be able to apply Newton's laws of motion;
- Understand and be able to apply other basic dynamics concepts – the Work-Energy principle, Impulse – Momentum principle and the coefficient of restitution;
- Learn to solve dynamics problems choosing an appropriate solution strategy;
- Attain an introduction to basic machine parts such as pulleys and mass-spring systems

Module	Content	Lecture
Module 1	Importance of Mechanics in engineering. Introduction to Statics; Concept of Particle and Rigid Body; Types of forces: collinear, concurrent, parallel, concentrated, distributed; Vector and scalar quantities; Force as a vector; Transmissibility of a force.	3
	Introduction to Vector Algebra; Parallelogram law; Lami's theorem; Co-ordinate representation of vector; Cross product and Dot product and their applications.	3
	Two dimensional force system; Resolution of forces; Moment; Varignon's theorem; Couple; Resolution of a coplanar force by its equivalent force-couple system; Resultant of forces	4
Module 2	Concept and Equilibrium of forces; Free body concept and diagram; Equations of equilibrium, plane frames and trusses.	4
	Concept of Friction; Coulomb friction; Angle of Repose; Coefficient of friction	4
Module 3	Distributed Force: Centroid and Centre of Gravity; Centroids, circular sector, quadrilateral, composite areas consisting of different figures.	4

	Moments of inertia: Parallel axis theorem; Perpendicular axis theorem; Mass moment of inertia of symmetrical bodies e.g. cylinder, sphere, cone etc.	4
	Concept of virtual work and energy; Concept of simple stresses and strains; Torsion; Concept of fluid statics	4
Module 4	Introduction to Dynamics: Kinematics and Kinetics; Newton's laws of motion; Law of gravitation & acceleration due to gravity; Rectilinear motion of particles; determination of position, velocity and acceleration under uniform and non-uniformly accelerated rectilinear motion; construction of x-t, v-t and a-t graphs.	4

Books Recommended:

1. Engineering Mechanics [Vol I & II] by Meriam & Kraige – Wiley India
2. Engineering Mechanics: Statics & Dynamics by R.C. Hibbeler – Pearson
3. Vector Mechanics for Engineers [Vol I & II] by F. P. Beer and E. R. Johnston, TMH
4. Engineering Mechanics by Timoshenko, Young and Rao – TMH

ESC	ECEUGES01	Basic Electronics Engineering (for CEN,MEN & CSE)	3L-0T-0P	3 Credits
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Course Outcomes:

Module	Content	Lecture
Module 1	Semiconductor Basics: Energy band theory, Fermi levels, Conductors, Semiconductors and Insulators: electrical properties, Semiconductors: intrinsic and extrinsic, P-type and N-type semiconductors; electrical conduction phenomenon, drift and diffusion carriers, mass action law.	7
Module 2	Rectifying Devices: Formation of P-N junction, formation of depletion zone, Junction capacitance-I characteristics, Zener breakdown, Avalanche breakdown. Linear piecewise model; rectifiers: half wave, full wave, ripple factor, efficiency, Clipper and Clamper circuits	6
Module 3	Transistors: Formation of PNP / NPN junctions, principle of operation, configurations, transistor characteristics. Biasing and Bias stability: small signal low frequency operation of transistors; equivalent circuits h parameters, Transistors as amplifier: voltage gain, current gain, input impedance and output impedance, Decibel power.	7

Module 4	Field Effect Transistor: Construction and characteristics of JFET and MOSFET characteristics; depletion and enhancement type, FET small signal model.	4
Module 5	Feed Back Amplifier: Block diagram, properties, positive and negative feedback, loop gain, topologies of feedback amplifier; effect of feedback on gain, output impedance, input impedance, sensitivities(qualitative), bandwidth stability	4
Module 6	Operational Amplifier: Introduction to integrated circuits, operational amplifier and its terminal properties; concept of virtual earth, Gain-frequency and Slew rate; inverting and non-inverting mode of operation, voltage summing, difference, voltage follower, integrator, and differentiator.	4
Module 7	Electronic Instruments: Principle of operation of CRO; Electron ballistics and electron beam deflection; Concept of time base; Measurement of voltage, and frequency.	4

Text Books:

1. Rakshit & Chattopadhyay, Foundation of Electronics, New Age
2. Cathey, Electronic Devices and Circuits, Shaum series, TMH
3. Boylestead & Nashlesky, Electronic Devices and Circuits, Pearson
4. Millman and Halkias, Integrated Electronics, TMH

BSC	MATUGBS01	Engineering Mathematics I	4L-0T-0P	4 Credits
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Course Outcomes:

Module	Content	Lecture
Module 1	Basics of sequence and series ; Power series, Limit, Continuity.	14
Module 2	Differentiation, Mean value theorems and its application; Taylor's theorem, Maclaurin's infinite series; Maxima and minima; L Hospital's rule.	12
Module 3	Reduction formulae, Beta and Gama functions.	2
Module 4	Lines and planes, Polar coordinates, Quadric surfaces, Volume, Area, length.	10

Module 5	Continuity, Differentiability of vector functions, Arc length; Curvature, Torsion, Serret-Frenet formulas, Double, triple integrals, Jacobian .	10
Module 6	Green theorem, Gauss theorem and Stokes Theorems and its application.	6

References:

1. Advanced Engineering Mathematics : Erwin Kreyszig
2. Advanced Engineering Mathematics : R.K. Jain & S. R. K lyengar
3. Advanced Engineering Mathematics : C. R. Wylle & L. C. Barrett
4. Differential & Integral Calculus : N. Plskunov

BSC	PHYUGBS01	Engineering Physics	3L-0T-0P	3 Credits
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Course Outcomes:

Module	Content	Lecture
Module 1	Mechanics Classical mechanics: scalars and vectors, vector multiplication, central force, mechanics of system of particles, elastic properties, rotational motion, fluid dynamics: viscosity, Stoke's law, streamline flow, equation of continuity, Reynold's number, Bernoulli's theorem Quantum mechanics: Photoelectric effect, de-Broglie's hypothesis, matter wave, Hysenberg's uncertainty principle, wave function, Schrodinger equation and simple problems Statistical mechanics: Necessity of statistical mechanics, Maxwell-Boltzman, Bose-Einstein and Fermi-Dirac distribution formula	12

Module 2	Optics Huygen's principle, Interference of light, Young's double-slit experiment, Newton's ring; Diffraction: Fresnel and Fraunhofer class, Fresnel's half-period zones, zone plate, Fraunhofer diffraction due to single slit and plane transmission grating (elementary theory); Polarization: plane, circular and elliptically polarized light, Brewster's law, Polaroid, optical activity. Coherence length and time; Einstein's A and B coefficients; spontaneous and induced emissions, condition for laser action, population inversion, He-Ne laser Optical Fiber, core and cladding; total internal reflection; optical fiber and waveguide; communication through optical fiber, energy loss, attenuation and dispersion	6
Module 3	Electrostatics & Electricity Coulomb's law, intensity and potential of point charge, Gauss's theorem and simple applications, electric-dipole, Electric displacement, capacitor, parallel plates and cylindrical, Thermoelectricity, Magnetic effects of currents, Self-inductance, Mutual inductance, Transformer Electric circuit elements and AC, DC circuit analysis.	6
Module 4	Solid State Physics Crystalline nature of solid, diffraction of X-ray, Bragg's law, Mosley's law, explanation from Bohr's theory, Origin of the energy gap, band theory; metal, semiconductor and insulators; intrinsic and extrinsic semiconductors, dia, para and ferro magnetic materials, superconductivity	6
Module 5	Nuclear Physics Binding energy of nucleus, Binding energy curve and stability, Radioactivity, successive disintegration, radioactive equilibrium, radioactive dating, radioisotope and their uses, Nuclear transmutation, fission & fusion, nuclear reactor	6

Text Book:

ESC	CENUGES01	Engineering Graphics & Design	0L-1T-3P	2.5 Credits
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Course Outcomes:

Module	Content	Lecture
Module 1	Introduction to Engineering Graphics: Drawing instruments and accessories, BIS – SP 46. Use of plane scales, Diagonal Scales and Representative Fraction. Engineering Curves: Classification and application of Engineering Curves, Construction of Conics, Cycloid Curves, Involute and Spirals along with normal and tangent to each curve.	30
Module 2	Projections of Points and Lines: Introduction to principal planes of projections, Projections of the points located in same quadrant and different quadrants, Projections of line with its inclination to one reference plane and with two reference planes. True length and inclination with the reference planes. Projections of Planes: Projections of planes (polygons, circle and ellipse) with its inclination to one reference plane and with two reference planes, Concept of auxiliary plane method for projections of the plane	30
Module 3	Projections of Solids and Section of Solids: Classification of solids. Projections of solids (Cylinder, Cone, Pyramid and Prism) along with frustum with its inclination to one reference plane and with two reference planes. Section of such solids and the true shape of the section	20
Module 4	Drawing practice: Drawing practise using software like AUTO CAD	20

Reference Books:

1. A Text Book of Engineering Graphics by P.J.Shah S.Chand & Company Ltd., New Delhi.
2. Elementary Engineering Drawing by N.D.Bhatt Charotar Publishing House, Anand.
3. A text book of Engineering Drawing by R.K.Dhawan, S.Chand & Company Ltd., New Delhi.

ESC	ECEUGES02	Basic Electronics Engineering Lab (for CEN,MEN & CSE)	0L-0T-3P	1.5 Credits
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Course Outcomes:

Module	Content	Lecture
Module 1	Familiarization of Electrical and Electronics Components	3
Module 2	Familiarization of Various Instruments like Power Supply, Digital Multimeter, Function	3
Module 3	Generator, CRO etc.	3
Module 4	Study of Junction Diode Characteristics	3
Module 5	Study of Zener Diode Characteristics	3
Module 6	Study of Clipping Circuits	3
Module 7	Study of Clamping Circuits	3
Module 8	Study of Rectifier Circuits	3
Module 9	Study of BJT Characteristics	3
Module 10	Study of FET Characteristics	3
Module 11	Study of fundamental characteristics of OP-AMP	3
Module 12	Determination of Slew rate and bandwidth of an OP-AMP.	3

BSC	PHYUGBS02	Engineering Physics Lab	0L-0T-3P	1.5 Credits
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Course Outcomes:

Module	Content	Lecture
Module 1	Measurements of length (or diameter) using vernier scale, slide caliper, screw gauge and travelling microscope.	3
Module 2	Determination of the radius of curvature of a spherical surface by using spherometer.	3
Module 3	Determination of moment of inertia of (a) a cylinder and (b) a rectangular solid bar.	3

Module 4	To determine the focal length of a concave lens by combination method and hence to determine the refractive index of the material of the lens by measuring the radii of curvature of both lenses	3
Module 5	Determination of the average resistance per unit length of the meter bridge wire by Carey-Foster's method and hence to determine an unknown resistance	3
Module 6	Determination of the horizontal component of the earth's magnetic field and the magnetic moment of a magnet by employing magnetometers	3
Module 7	Determination of Young's Modulus of elasticity of a material of a bar by the method of flexure.	3
Module 8	Determination of rigidity modulus of a material of a wire by static method.	3
Module 9	Determination of rigidity modulus of a material of a wire by dynamic method.	3
Module 10	Determination of unknown frequency of a tuning fork by using a sonometer.	3

MC	UCCUGAU01	Elementary Arabic & Islamic Studies	4L-0T-0P	0 Credits
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Course Outcomes:

SYLLABUS

TEXT BOOKS NOT MENTIONED

Module 4	Safety, Responsibilities and Rights : Safety definition, Safety and risk, Risk analysis, Assessment of safety and risk, Conflict of interests, Occupational crime, Human rights, Employee rights, Whistle blowing, Intellectual property rights.	5
Module 5	Global Issues : Globalization, Multinational corporations, Environmental ethics, Computer ethics, Engineers as managers, Engineers as advisors in planning and policy making, Moral leadership, Codes of ethics.	5

Books:

1. A Textbook of Professional Ethics and Human Values, R.S. Naagarazan, New Age International Publishers
2. Blending the best of the East & West, Dr. Subir Chowdhury, EXCEL
3. Ethics & Mgmt. & Indian Ethos, Ghosh, VIKAS
4. Business Ethics, Pherwani, EPH
5. Ethics, Indian Ethos & Mgmt., Balachandran, Raja, Nair, Shroff Publishers Business Ethics: concept and cases, Velasquez, Pearson.

OEC402	Open Elective course Will be offered by others department	3L:0T:0P	3credits
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MENUGPR04	Preproject-IV	6	3credits
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Developed components in project –III will be assembled for demonstrating certain principles of mechanical Engineering.